

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 1.5 RESPIRATION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit on Respiration. Return to the sealed thermos of uji that Achieng, a Form 4 student in Kisumu, opened at school. Use the evidence you collected in your investigations, your Driving Question Board and your model to answer each section in the space provided below it. Write in full sentences, use correct science words, and name the evidence that supports each idea. Where a question asks for an equation, write it clearly. Take your time, and remember to link every answer back to the thermos of uji.

Part 1: What is happening inside the sealed thermos?

Achieng poured fermented uji from a sealed thermos flask. She saw tiny bubbles and noticed that the uji smelled more sour than it did two hours earlier. No heat and no new ingredients were added. A jar of fresh uji wa kawaida (not fermented) did not change. (a) Explain what is producing the bubbles and the sour taste. (b) Use evidence from the yeast and lime water investigation (Flask A and Flask B) to support your explanation. (c) Explain why the process can continue inside a sealed flask.

Write your answer here:

Part 2: Aerobic and anaerobic respiration compared

Amina, a sprinter in Nairobi, gets energy from the ugali she eats. (a) Write the word equation for aerobic respiration and state where in the cell it mainly happens. (b) Explain why aerobic respiration alone cannot explain the bubbles in Achieng's sealed thermos. (c) Write the word equation or chemical equation for what happens in Amina's muscle cells when oxygen supply is not enough, and say what Amina feels. (d) Compare the amount of ATP released by the two types of respiration from the same amount of glucose, and say which releases more.

Write your answer here:

Part 3: Evidence for sourness, substrates and RQ

(a) In the universal indicator test, fresh uji, fermented uji from the sealed thermos, mala and a distilled water control were compared. Describe what the results showed for the uji samples and the distilled water, and explain what they tell you about the cause of sourness. Say what the mala result suggests. (b) Name the three main respiratory substrates, give a Kenyan food example of each, and state the RQ values you learned. (c) Write the formula for RQ. (d) The RQ is worked out for cells that are using oxygen. Explain what happens to the RQ value if anaerobic respiration is also contributing CO₂ in addition to aerobic respiration, and what that tells you about the type of respiration.

Write your answer here:

Part 4: Why does uji ferment at different rates?

Achieng's uji was kept in a thermos. Imagine the same uji in Mombasa (hot, humid coast) and in Kericho (cool highlands). (a) Predict where it would ferment faster and explain why, using what you found in the yeast investigation. (b) Name two other factors besides temperature that affect the rate of respiration and describe their effect with evidence from the class results. (c) Explain why a farmer in Kenya would keep stored maize cool and dry.

Write your answer here:

Part 5: Using respiration to solve community problems

Answer the driving question: How do living organisms, from the yeast in our uji to the cells in our own muscles, release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities? (a) Write a three-sentence answer. (b) Give TWO Kenyan examples of anaerobic respiration used in the economy, naming the organism or product involved and the by-product that is useful. (c) Explain why brewing alcoholic drinks is not a responsible application of what you learned.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the anchoring phenomenon	Clearly explains that yeast (and bacteria) respire anaerobically in the sealed thermos, linking bubbles to CO ₂ and sourness to acid products, and uses both the Flask A/Flask B and pH evidence accurately.	Explains that microorganisms cause the bubbles and sourness but uses evidence from only one investigation or with small gaps.	States that something changes in the uji but does not link it to respiration or to evidence.
Aerobic and anaerobic respiration equations and comparison	Writes correct word or chemical equations for aerobic respiration, alcoholic fermentation and lactic acid fermentation, and correctly compares oxygen need and ATP released.	Writes most equations correctly and makes a basic comparison with a minor error or omission.	Equations are missing or incorrect and the comparison is unclear.
Substrates, RQ and rate factors	Correctly names substrates with Kenyan examples and RQ values, uses the RQ	Gives most substrate, RQ and factor information correctly, with limited use of	Gives few correct facts about substrates, RQ or rate factors and does not use data.

	formula, interprets RQ above 1.0, and explains temperature and glucose concentration effects using class data.	data.	
Application to Kenyan economy and community	Gives two accurate Kenyan applications naming organism, product and benefit, and explains the responsible-use caution on brewing.	Gives two applications with some missing detail, or a limited explanation of the caution.	Gives one or no application or the applications are incorrect.
Scientific communication and use of evidence	Writes in clear, organised sentences, uses scientific terms correctly, and links every part back to the uji thermos.	Writing is mostly clear with mostly correct terms and some links to the thermos.	Writing is unclear, terms are misused and links to the thermos are missing.

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Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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